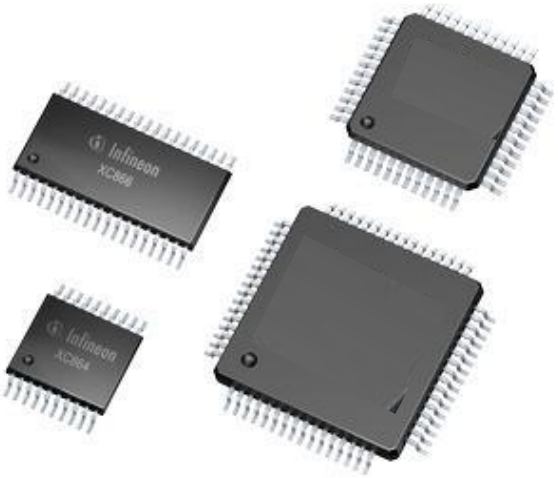
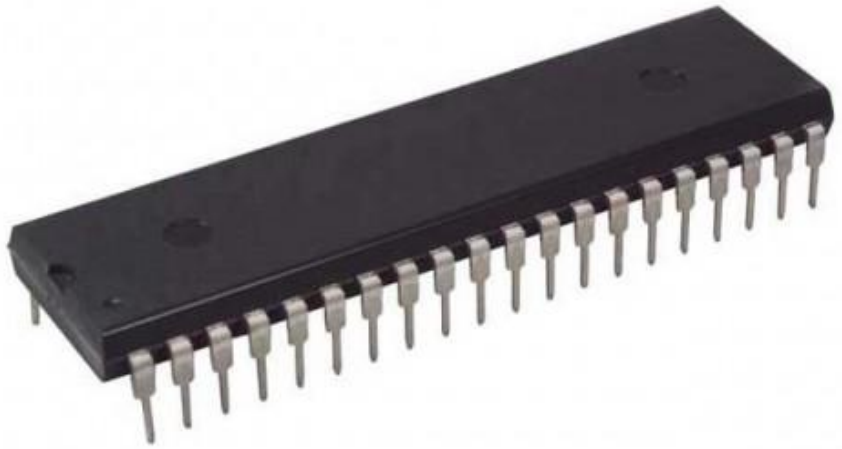
8051 Microcontroller





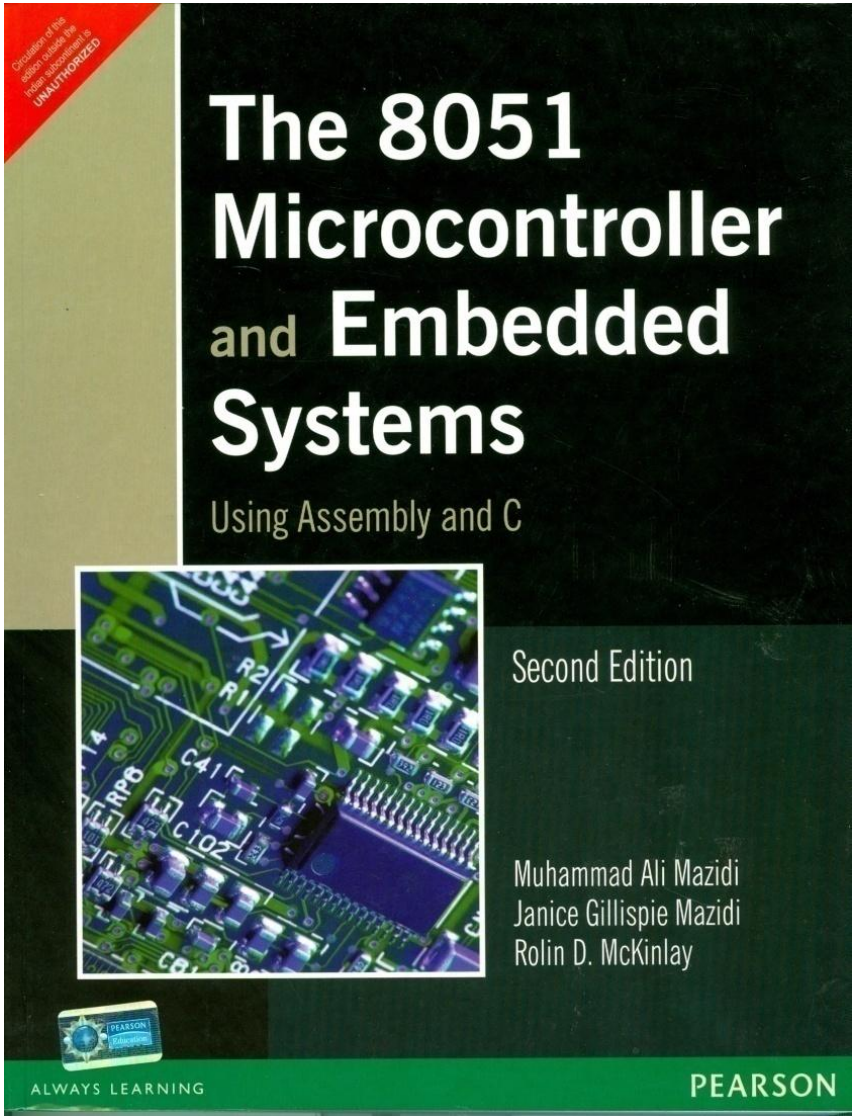
Course Objectives:

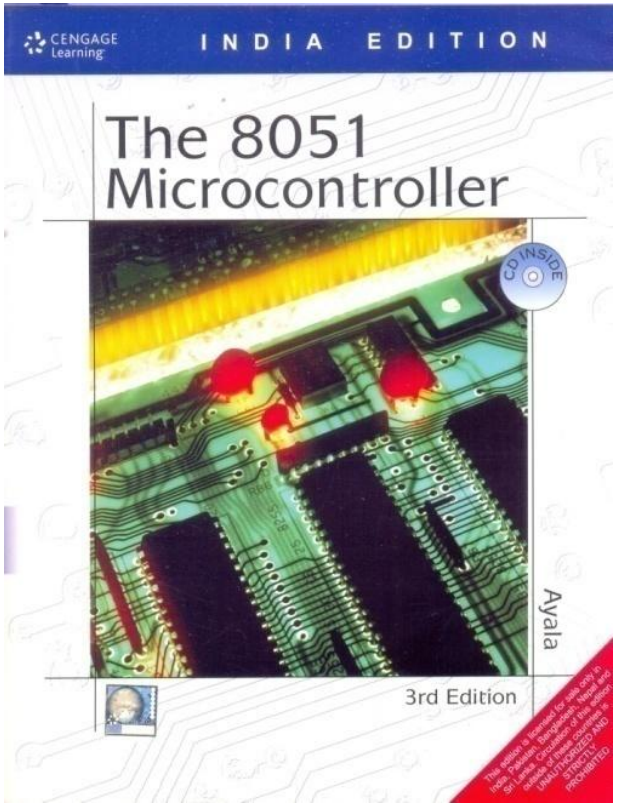
· To understand architecture and features of typical Microcontroller.
· To understand need of microcontrollers in real life applications.
· To learn interfacing of real world peripheral devices
· To study various hardware and software tools for developing applications.

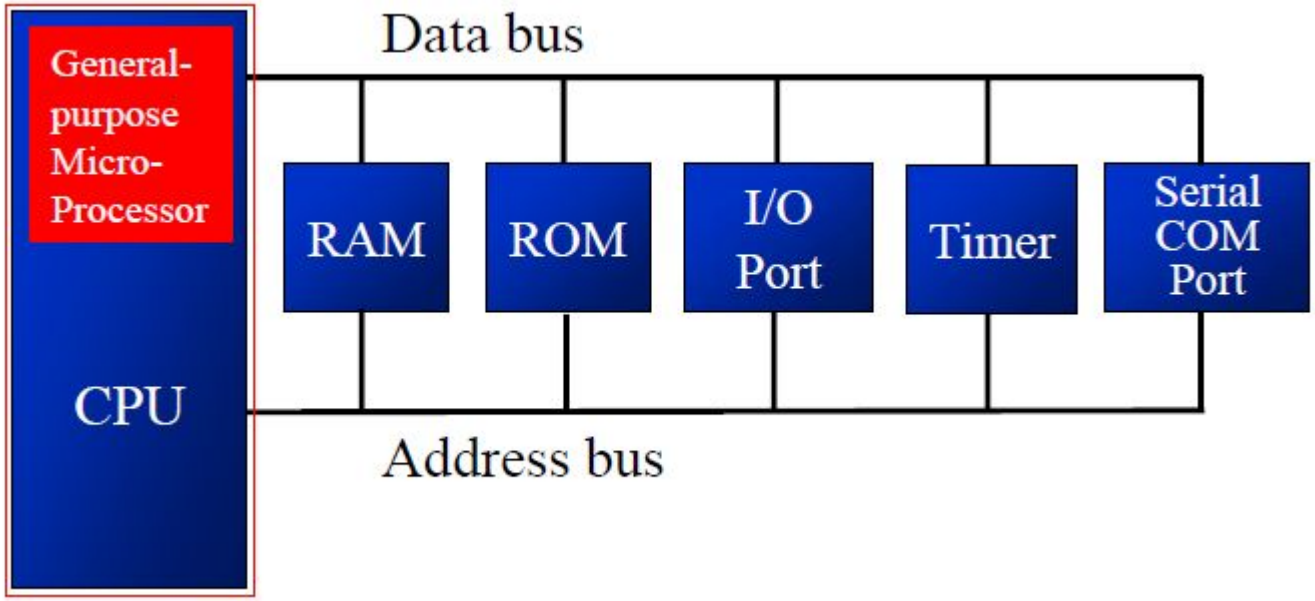
Course Outcomes:

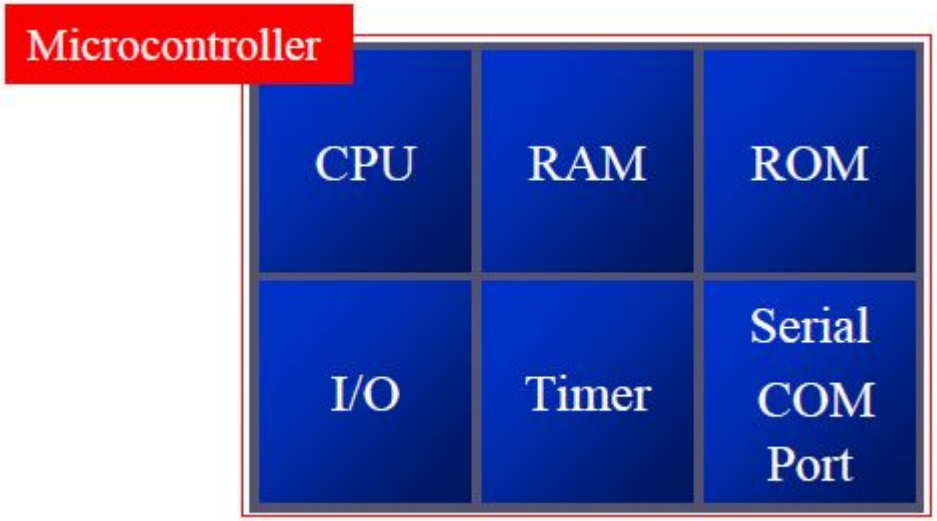
On completion of the course, student will be able to

1) Learn importance of microcontroller in designing embedded application.
2) Learn use of hardware and software tools.
3) Develop interfacing to real world devices.


Circulation of this edition outside the Indian subcontinent is UNAUTHORIZED
The 8051 Microcontroller and Embedded Systems
Using Assembly and C
Second Edition
Muhammad Ali Mazidi
Janice Gillispie Mazidi
Rolin D. McKinlay
ALWAYS LEARNING
PEARSON


CENGAGE Learning
INDIA EDITION
The 8051 Microcontroller
CD INSIDE
Ayala
3rd Edition


General-purpose Micro-Processor
CPU
Data bus
RAM
ROM
I/O Port
Timer
Serial COM Port
Address bus

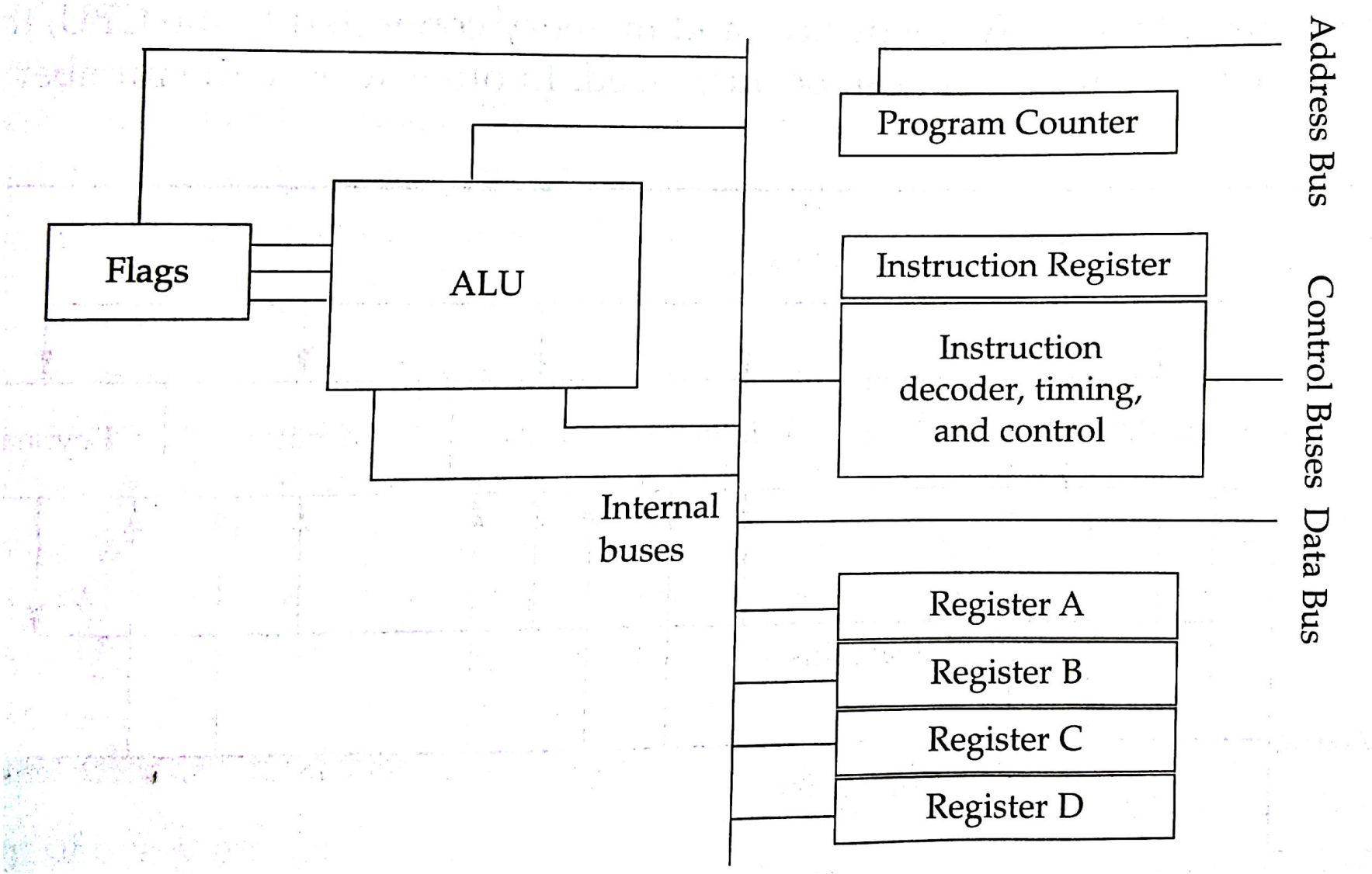

Microcontroller
CPU
RAM
ROM
I/O
Timer
Serial COM Port

Advantages of microprocessor

- Designer can decide on amount RAM,ROM,I/O ports needed to fit task at hand

Advantages of Microcontroller

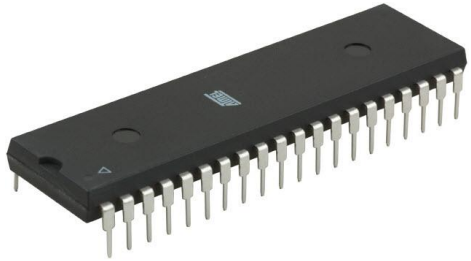
- Less space requirements
- Low power consumption


Address Bus
Control Buses
Data Bus
Program Counter
Flags
ALU
Instruction Register
Instruction decoder, timing, and control
Internal buses
Register A
Register B
Register C
Register D

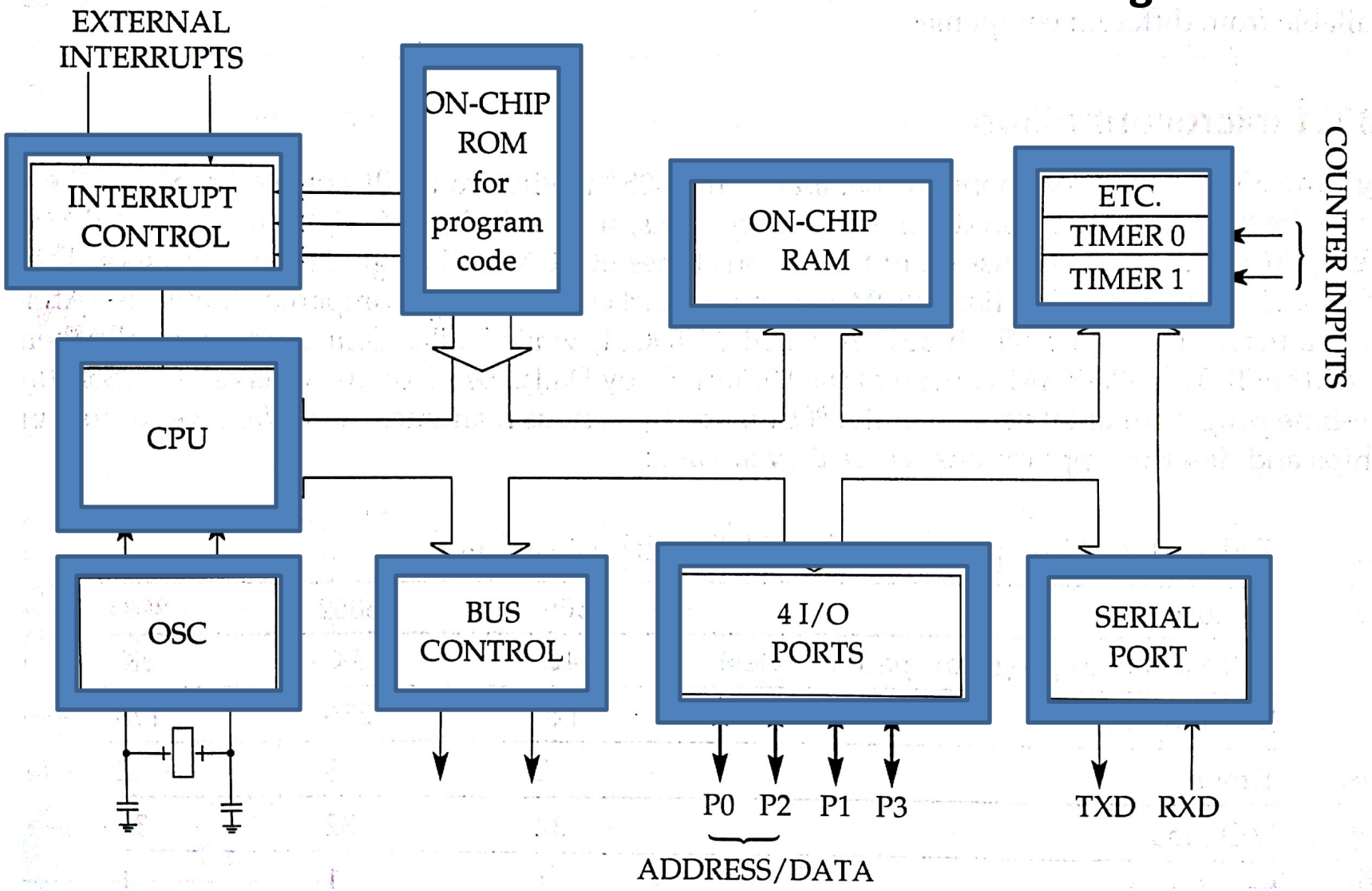
Criteria in Choosing a Microcontroller

Speed	• What is the highest speed that microcontroller can support
memory	• Amount of RAM and ROM on chip
I/O pin	• No of I/O pins available for
Power consumption	• Critical for battery powered product
Cost per unit	• Important in term of final cost of package)
packaging	• DIP(Dual in line package) QFP(quad flat package
Timer/ serial Availability	• Delay, Serial communication • Availability of
ty of development	assembler, compiler,etc

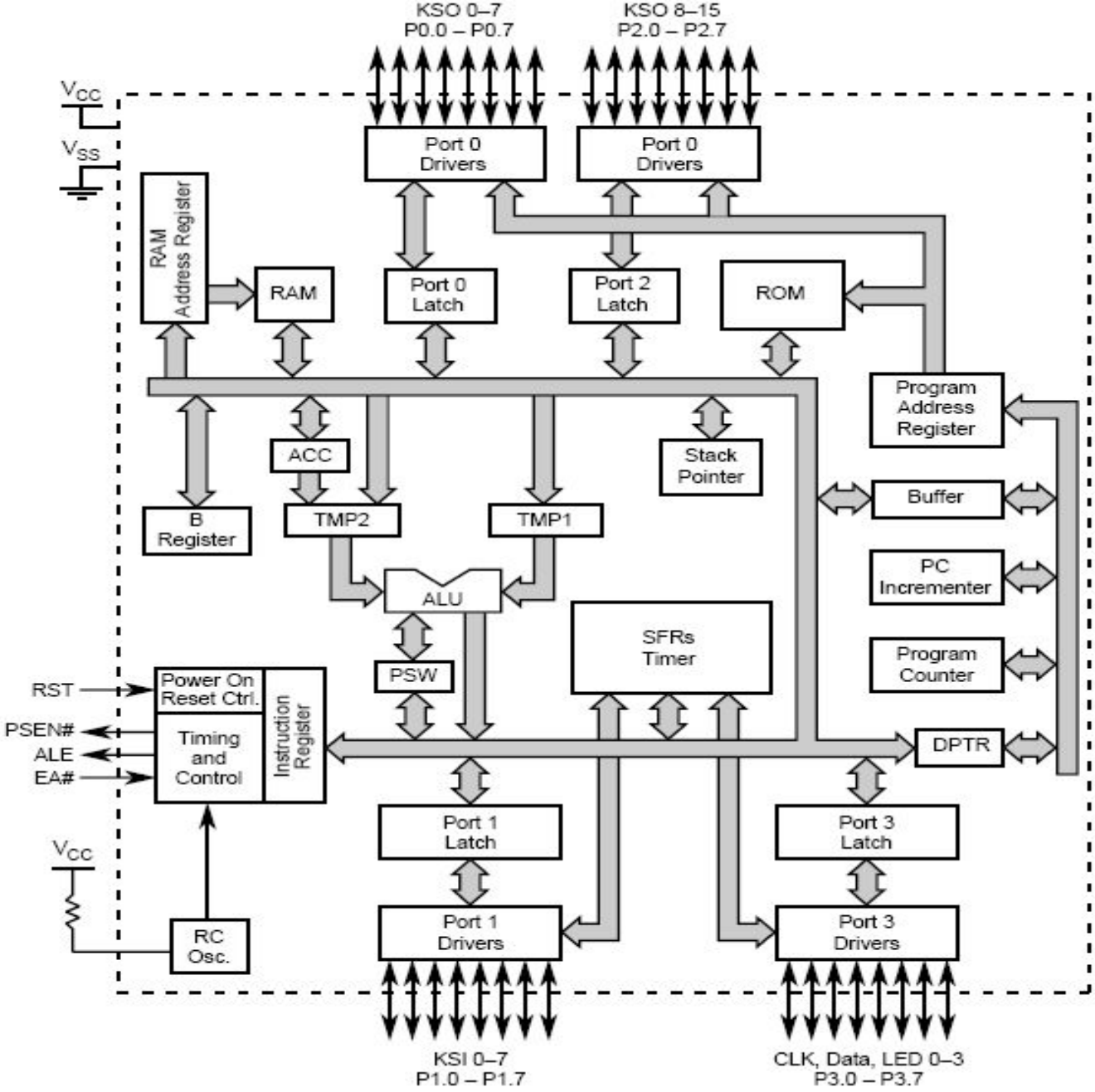




8051 Block Diagram



8051 Internal Architecture



- 8051 MC
  - 8 bit microcontroller
- RAM
  - 128 byt
- ROM
  - 4 K byt
- Timer
  - 2-16
- Serial port
  - 1 se
- I/O port
  - 4 I/O
- Data Bus
  - 8 bit
- Address
  - 16
- 8051 mc
  - 8 bit mic



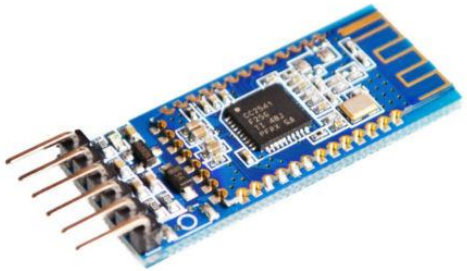
Other 8051 features

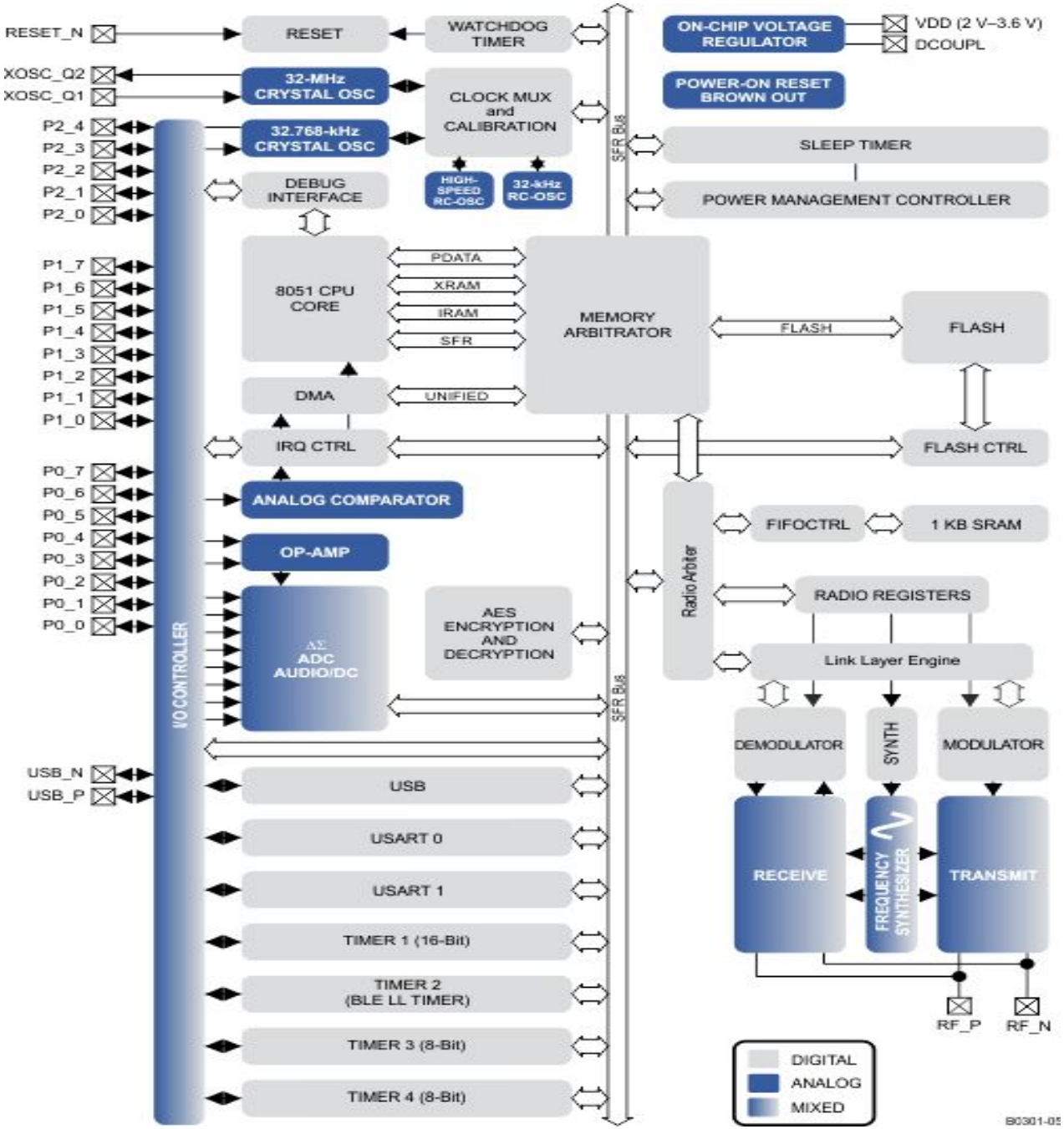
- only 1 On chip oscillator (external crystal)
- 6 interrupt sources (2 external , 3 internal, Reset)
- Upto 64K external code (program) memory(only read)PSEN
- Upto 64K external data memory(can be read and write) by RD,WR
- Code memory is selectable by EA (internal or external)
- We may have External memory as data and code

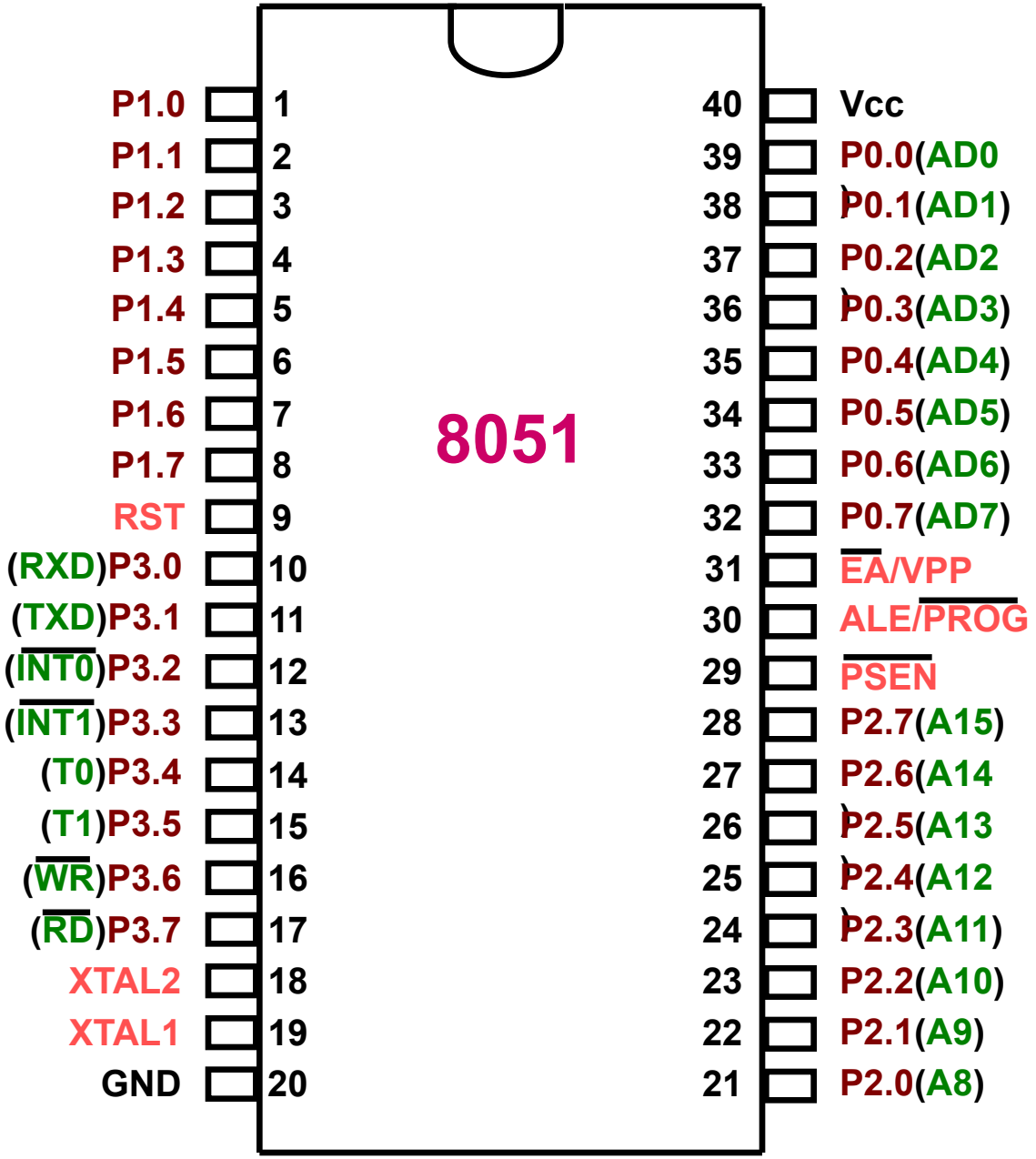
Comparison of 8051 Family Members

Feature	8051	8052	8031
ROM (on-chip program space in bytes)	4K	8K	0K
RAM (bytes)	128	256	128
Timers	2	3	2
I/O pins	32	32	32
Serial port	1	1	1
Interrupt sources	6	8	6

Industry
8051
Micro-con
troller



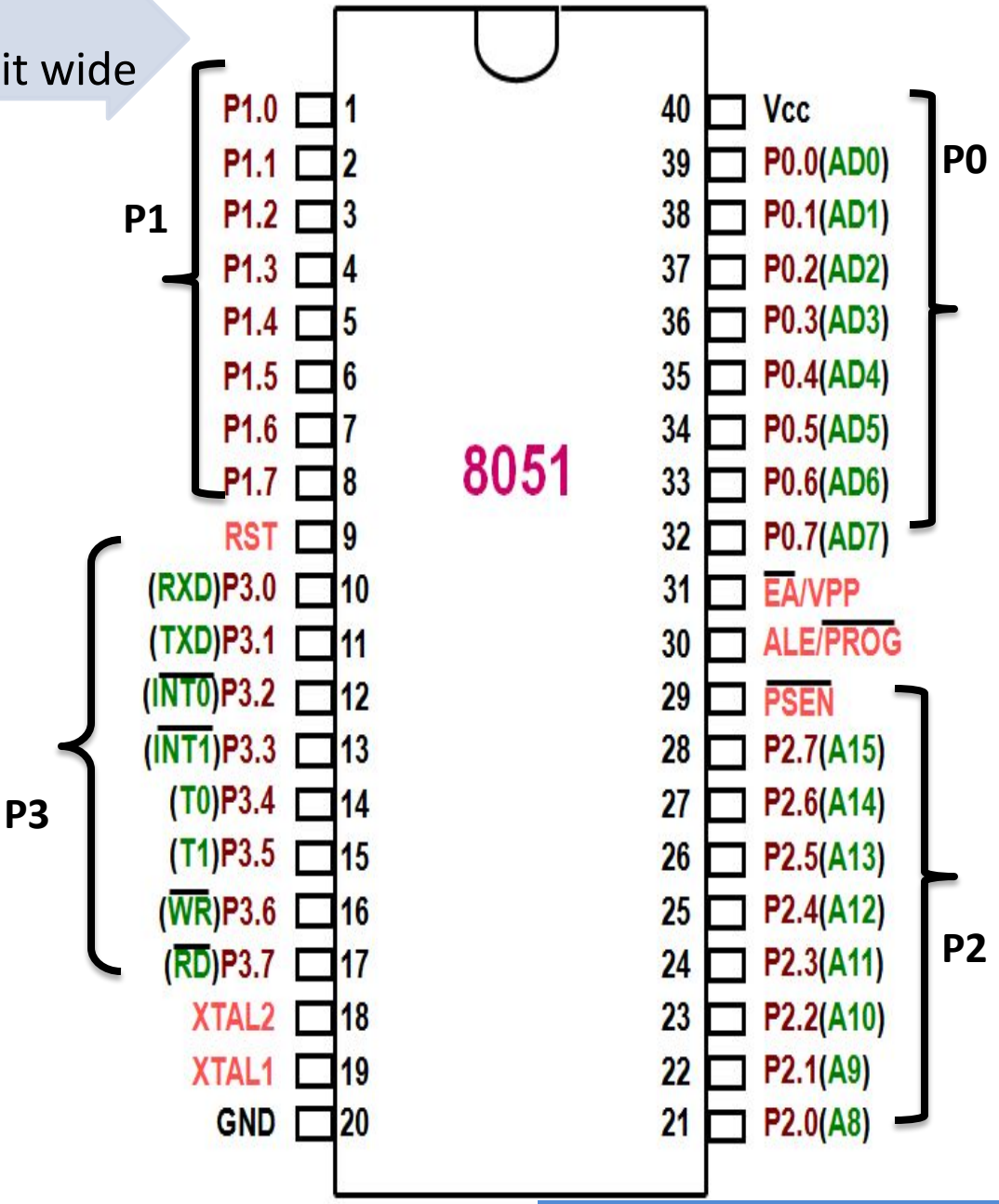


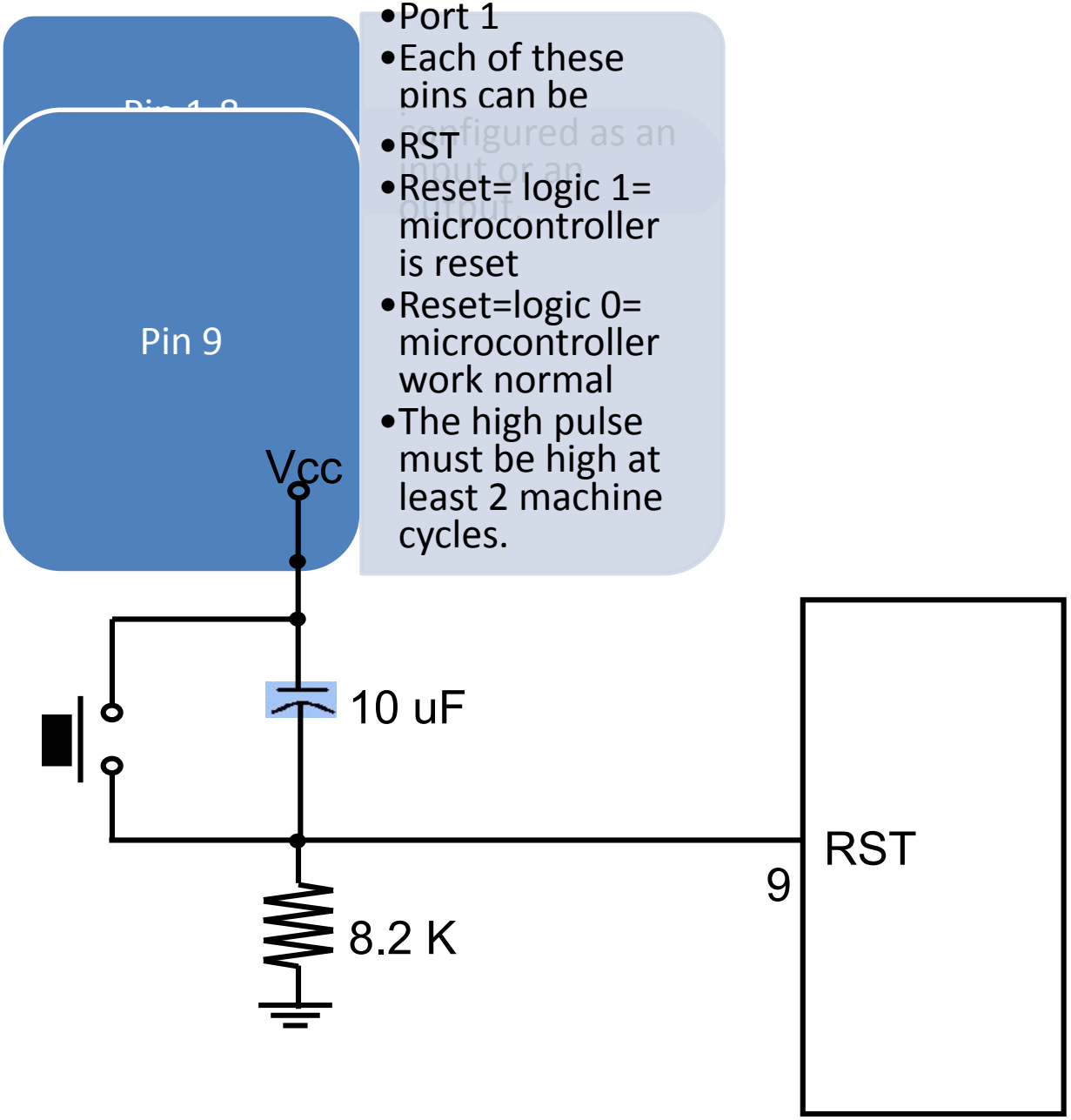

8051
P1.0 1
P1.1 2
P1.2 3
P1.3 4
P1.4 5
P1.5 6
P1.6 7
P1.7 8
RST 9
(RXD)P3.0 10
(TXD)P3.1 11
(INT0)P3.2 12
(INT1)P3.3 13
(T0)P3.4 14
(T1)P3.5 15
(WR)P3.6 16
(RD)P3.7 17
XTAL2 18
XTAL1 19
GND 20
40 Vcc
39 P0.0(AD0
38 P0.1(AD1)
37 P0.2(AD2
36 P0.3(AD3)
35 P0.4(AD4)
34 P0.5(AD5)
33 P0.6(AD6)
32 P0.7(AD7)
31 EA/VPP
30 ALE/PROG
29 PSEN
28 P2.7(A15)
27 P2.6(A14
26 P2.5(A13
25 P2.4(A12
24 P2.3(A11)
23 P2.2(A10)
22 P2.1(A9)
21 P2.0(A8)

I/O port

- 4 I/O port
- Each of 8 bit wide

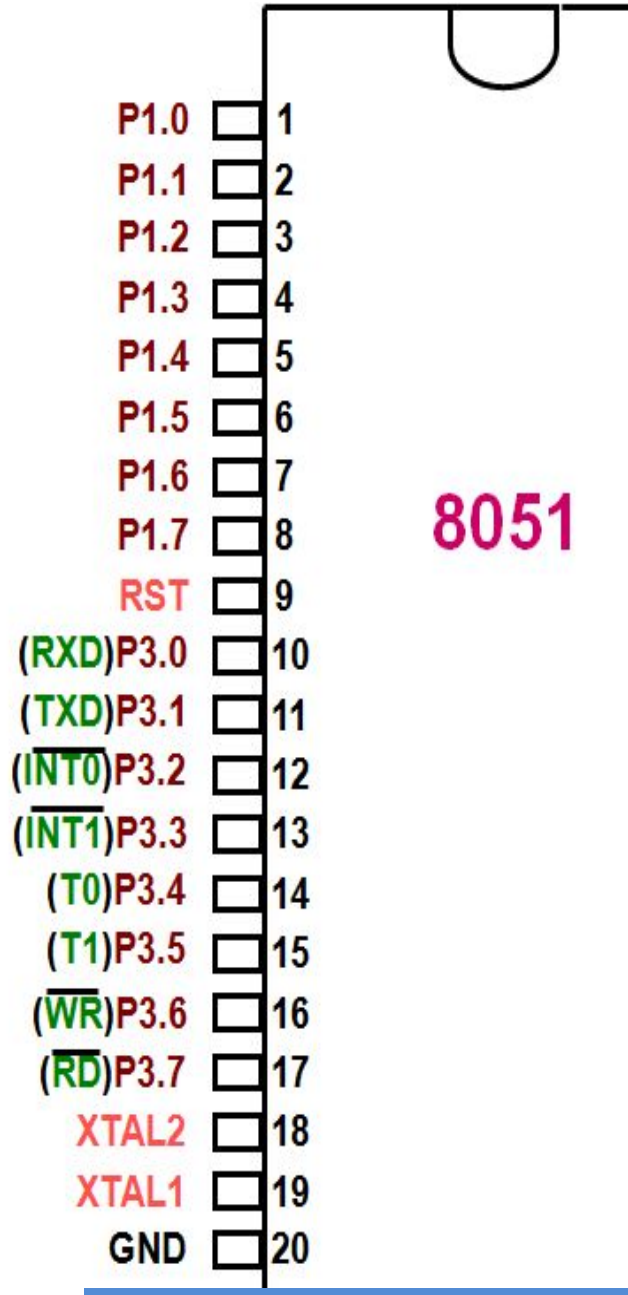
•4 I/O ports
- P0
  - P0.0
  - P0.1
  - P0.2
  - P0.3
  - P0.4
  - P0.5
  - P0.6
  - P0.7
- P1
- P2
- P3





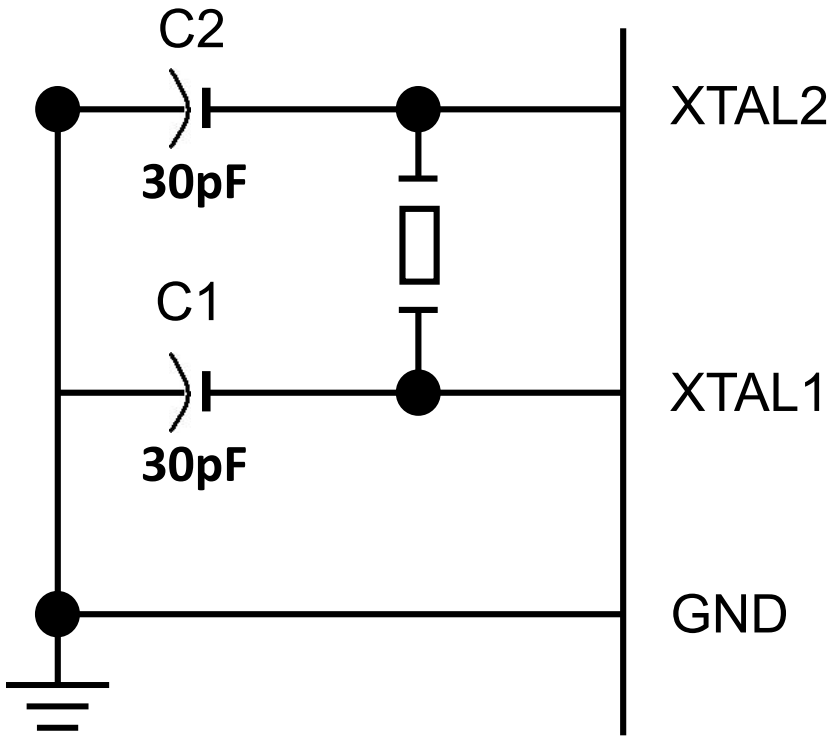
- Port 1
- Each of these pins can be configured as an input or an output
- RST
- Reset= logic 1= microcontroller is reset
- Reset=logic 0= microcontroller work normal
- The high pulse must be high at least 2 machine cycles.

Pin 10	• RXD • Use for serial communication
Pin 11	• TXD • Use for serial communication
Pin 12	• $\overline{\text{INT0}}$ • External interrupt 0
Pin 13	• $\overline{\text{INT1}}$ • External interrupt 1
Pin 14	• T0 • Timer/ counter 0
Pin 15	• T1 • Timer/ Counter 1

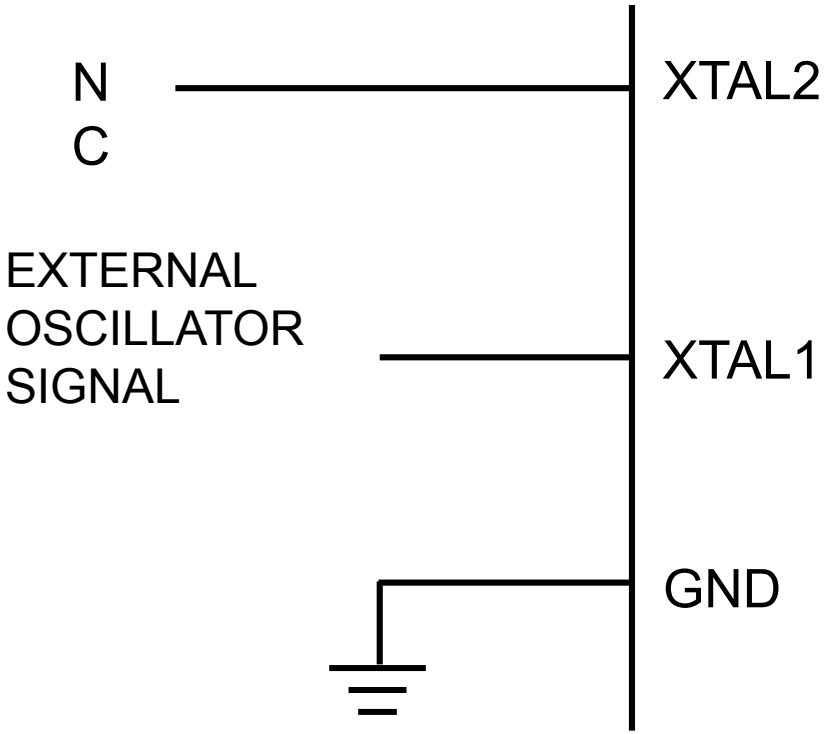


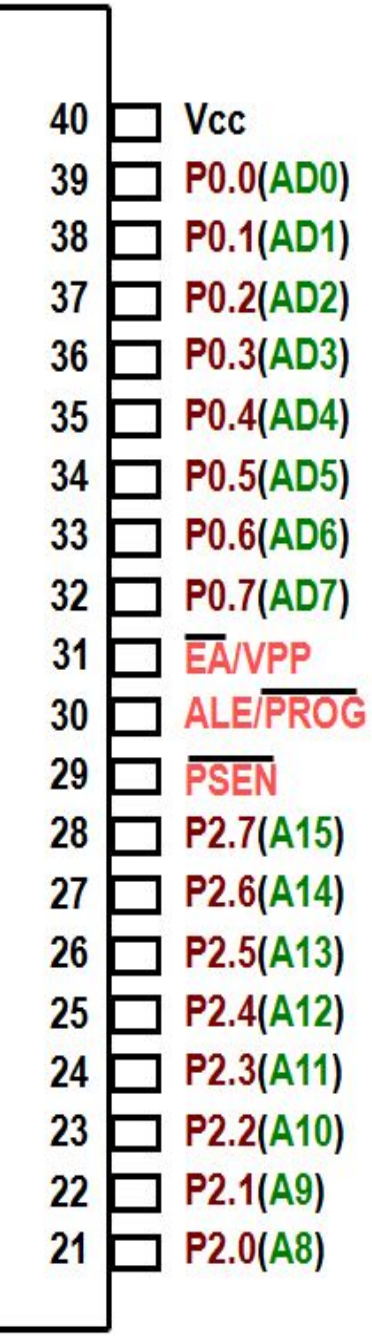
Pin 16	•$\overline{WR}$ •Write
Pin 17	•$\overline{RD}$ •Read
Pin 18	•XTAL2 •provide external clock.
Pin 19	•XTAL1 •provide external clock.

- Using a quartz crystal oscillator
- We can observe the frequency on the XTAL2 pin.



- Using a TTL oscillator
- XTAL2 is unconnected.



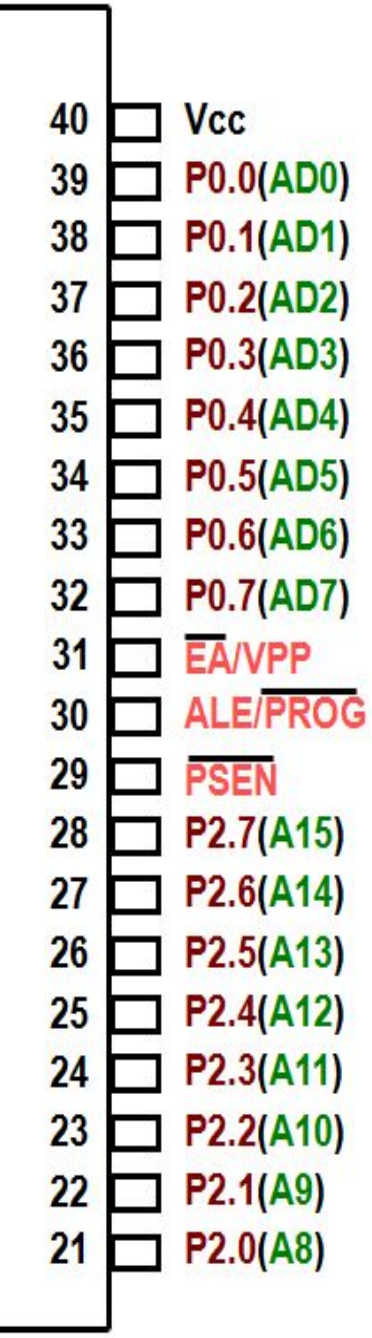


Pin 20

- GND
- ground

Pin 21-28

- Port 2/ A8-A15
- If there is no intention to use external memory then these port pins are configured as general inputs/outputs. In case external memory is used, the higher address byte, i.e. addresses A8-A15 will appear on this port.



Pin 29

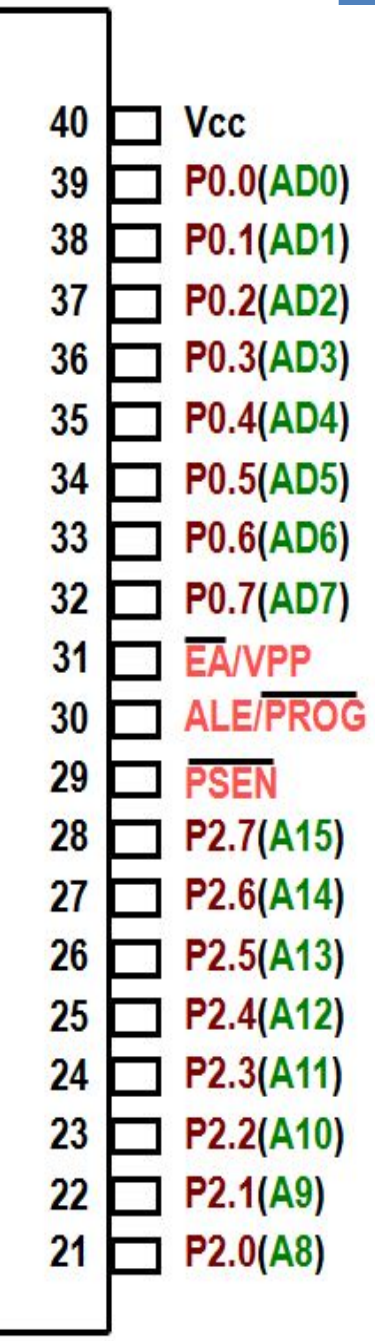
- PSEN
- Program store enable
- In 8031 based system in which external ROM holds the program code, this pin connected to OE pin of ROM

Pin 30

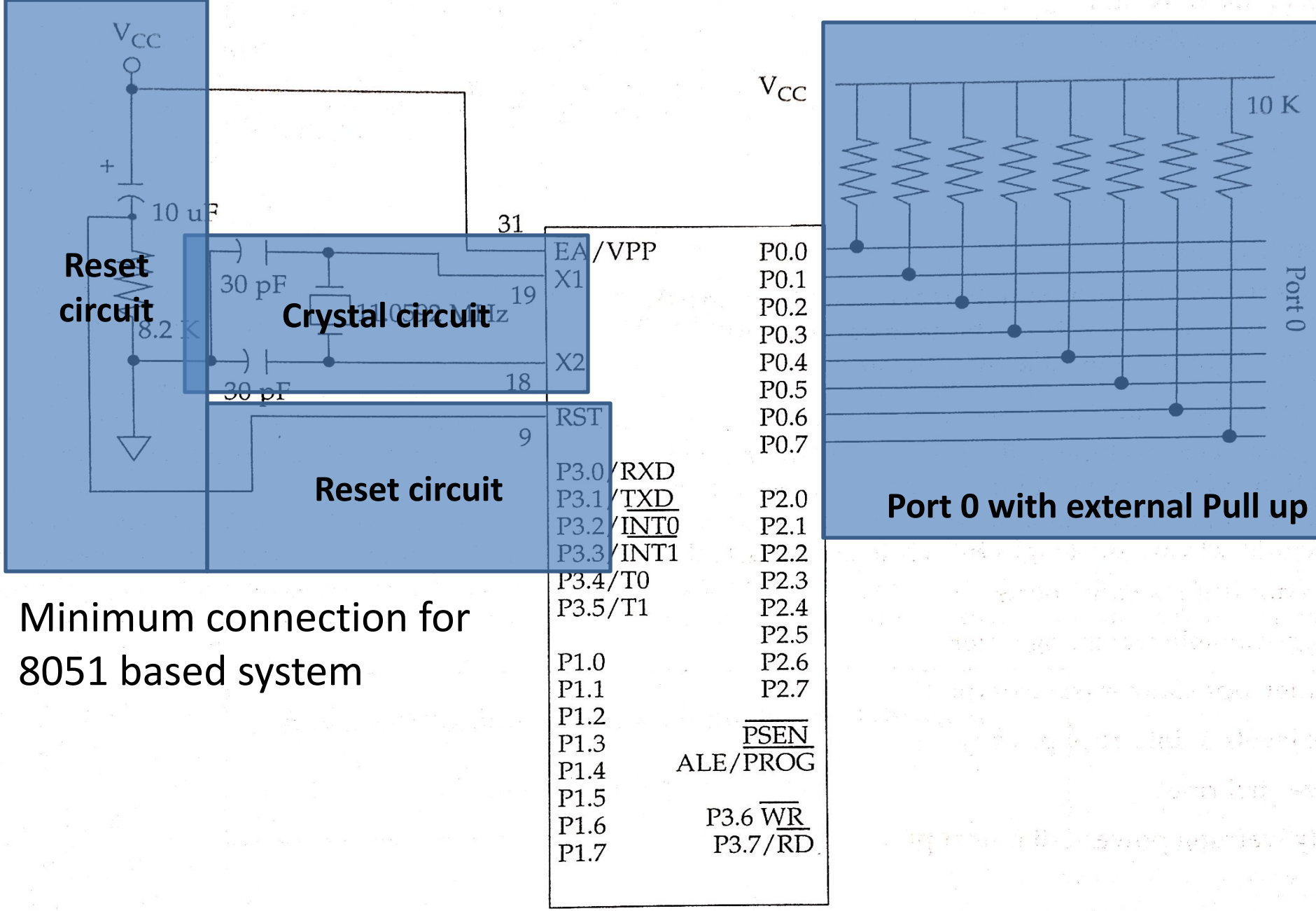
- ALE
- Used to demultiplex address and data

Pin 31

- EA
- External access
- On chip ROM store programs then EA=VCC
- External ROM store programs then EA= GND



Pin 32-39	•Port 0 / AD0-AD7 •Similar to P2, if external memory is not used, these pins can be used as general inputs/outputs. Otherwise, P0 is configured as address output (A0-A7) when the ALE pin is driven high (1) or as data output (Data Bus) when the ALE pin is driven low (0).
Pin 40	•VCC •Power supply



Minimum connection for 8051 based system